

Unit II: Question Bank

Limits, Continuity, Differentiability and Series of Functions of Two Variables

Section A: Multiple Choice Questions

1. If $f(x, y) = x^2 + y^2$, then $\lim_{(x, y) \rightarrow (0, 0)} f(x, y)$ is:

- A) 0
- B) 1
- C) ∞
- D) Does not exist

2. The function $f(x, y) = (x^2y)/(x^2 + y^2)$, for $(x, y) \neq (0, 0)$, and $f(0, 0) = 0$ is at $(0, 0)$:

- A) Discontinuous
- B) Continuous
- C) Differentiable but not continuous
- D) Undefined

3. For a function $f(x, y)$ to be continuous at (a, b) , which condition is necessary?

- A) $\lim f(x, y) = f(a, b)$
- B) $f(a, b) = 0$
- C) $f_x(a, b) = f_y(a, b)$
- D) Both partial derivatives must be zero

4. If $f(x, y) = x^2y + 3xy^2$, then $\partial f / \partial x$ is:

- A) $2xy + 3y^2$
- B) $x^2 + 6xy$
- C) $2xy + 3xy^2$
- D) $2x + y$

5. If $f(x, y) = e^{(xy)}$, then $\partial f / \partial y$ is:

- A) $xe^{(xy)}$
- B) $ye^{(xy)}$
- C) $xye^{(xy)}$
- D) $e^{(xy)}$

6. For $f(x, y) = x^3 + 3x^2y + 3xy^2 + y^3$, the value of $x f_x + y f_y$ is:

- A) $f(x, y)$
- B) $2f(x, y)$
- C) $3f(x, y)$
- D) $4f(x, y)$

7. Euler's theorem for a homogeneous function $f(x, y)$ of degree n states that:

- A) $x f_x + y f_y = f$
- B) $x f_x + y f_y = n f$
- C) $x f_x + y f_y = n$
- D) $f_x + f_y = n f$

8. The function $f(x, y) = x^2 + xy + y^2$ is homogeneous of degree:

- A) 1
- B) 2
- C) 3
- D) 0

9. If $z = f(x, y)$, the total differential dz is:

- A) $f_x dx + f_y dy$
- B) $f_x + f_y$
- C) $f_x dy + f_y dx$
- D) $dx + dy$

10. If $z = x^2y + \sin y$, then dz is:

- A) $(2xy)dx + (x^2 + \cos y)dy$
- B) $(x^2)dx + (2xy + \cos y)dy$
- C) $2x dx + \cos y dy$
- D) $x^2y dx + \sin y dy$

11. A sufficient condition commonly used for differentiability of $f(x, y)$ at a point is that:

- A) f is bounded
- B) f_x and f_y exist and are continuous in a neighborhood of the point
- C) $f_x = f_y = 0$
- D) f is homogeneous

12. If f_x and f_y exist at a point but are not continuous there, then f is:

- A) Always differentiable
- B) Always continuous
- C) Not necessarily differentiable
- D) Always homogeneous

13. The second-order Taylor expansion of $f(x, y)$ about (a, b) begins with:

- A) $f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b)$
- B) $f_x(a, b) + f_y(a, b)$
- C) $f(a, b)(x - a)(y - b)$
- D) $f(a, b) + x + y$

14. For $f(x, y)$, the quadratic terms in its Taylor expansion about (a, b) contain:

- A) f_x and f_y only

- B) f_{xx} , f_{xy} and f_{yy}
- C) f only
- D) No derivatives

15. The Maclaurin series is a Taylor series expanded about:

- A) (1,1)
- B) (a,b)
- C) (0,0)
- D) (∞, ∞)

16. The Maclaurin expansion of $e^{(x+y)}$, up to first-degree terms, is:

- A) $1 + x + y$
- B) $x + y$
- C) $1 + xy$
- D) $e + x + y$

17. For $f(x, y) = \sin(x + y)$, the first-degree Maclaurin approximation is:

- A) $x + y$
- B) $1 + x + y$
- C) xy
- D) $\sin x + \sin y$

18. If $f(x, y) = x^2 + y^2$, then the first partial derivatives at (1,2) are:

- A) $f_x = 1, f_y = 2$
- B) $f_x = 2, f_y = 4$
- C) $f_x = 2, f_y = 2$
- D) $f_x = 4, f_y = 2$

19. For a homogeneous function of degree n , Euler's theorem gives $x f_x + y f_y$ equal to:

- A) n
- B) $nf(x, y)$
- C) $f(x, y)/n$
- D) $x + y$

20. If a function is differentiable at a point, then it must be:

- A) Continuous at that point
- B) Discontinuous at that point
- C) Homogeneous
- D) Constant

Section B: Short Answer Questions

1. Evaluate $\lim_{x,y} \rightarrow (0,0), f(x,y) = [x^2y/(x^2 + y^2)]$.
2. Define the limit of a function of two variables at a point. Explain the conditions under which the limit exists, with a suitable example.
3. Define continuity of a function of two variables at a point (a,b) . Determine whether $f(x,y) = xy/(x^2 + y^2)$, for $(x,y) \neq (0,0)$, and $f(0,0) = 0$, is continuous at $(0,0)$.
4. Define the first-order partial derivatives of a function $z = f(x,y)$. Find f_x and f_y for $f(x,y) = x^3y^2 + 2x^2y + y^3$.
5. Find the first and second-order partial derivatives of $z = x^2y + xy^2 + e^{(xy)}$.
6. If $f(x,y) = \log(x^2 + y^2)$, find f_x, f_y, f_{xx} and f_{yy} .
7. If $z = x^2y + \sin(xy)$, find the total differential dz .
8. If $u = \sin^{-1}(x - y), x = 3t, y = 4t^3$, show that $\frac{du}{dt} = \frac{3}{\sqrt{1-t^2}}$.
9. If $z = u^2 + v^2, u = r\cos\theta, v = r\sin\theta$ find $\frac{\partial z}{\partial r}$ and $\frac{\partial z}{\partial \theta}$.
10. Define a homogeneous function. Verify Euler's theorem for $f(x,y) = x^3 + 3x^2y + 3xy^2 + y^3$.
11. State and prove Euler's theorem for a homogeneous function of degree n in two variables.
12. Verify Euler's theorem for $f(x,y) = (x^2 + y^2)/(x + y)$, and determine its degree of homogeneity.
13. State the Taylor series expansion of a function of two variables about the point (a,b) , up to second-order terms.
14. Expand $f(x,y) = e^{(x+y)}$ by Maclaurin's theorem, retaining terms up to the second degree.
15. Obtain the Maclaurin expansion of $f(x,y) = \sin(x + y)$ up to terms of degree three.

Section C: Long Answer Questions

1. If $u = r^n, r = \sqrt{x^2 + y^2}$, show that $xu_x + yu_y = nu$
2. Expand $f(x,y) = e^x \cos y$ in Taylor series near $(0,0)$ up to second order terms.
3. Expand $f(x,y) = \sin x \cos y$ about $(0,0)$ using Taylor's theorem up to third order term.
4. For $u = \ln(x^2 + y^2)$ verify that it is harmonic and find its harmonic conjugate.
5. For $u = x^m y^n$, verify Euler's theorem and find u_{xx} .
6. Evaluate the limit, if it exists
$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y}{x^4 + y^2}$$
7. If $z = x^3 y^2 + e^{xy} + \sin(x + y)$ find $\frac{\partial z}{\partial x}, \frac{\partial z}{\partial y}, \frac{\partial^2 z}{\partial x^2}, \frac{\partial^2 z}{\partial y^2}, \frac{\partial^2 z}{\partial x \partial y}$. Also verify whether $\frac{\partial^2 z}{\partial x \partial y} = \frac{\partial^2 z}{\partial y \partial x}$.
8. Find the total differential of $z = x^2 e^y + y^2 \ln x$.

9. Verify Euler's theorem for the homogeneous function $u = \frac{x^3+y^3}{x+y}$ and determine its degree of homogeneity. Further, verify the second-order Euler relation for the function.
10. Obtain the Maclaurin expansion of $f(x, y) = \ln(1 + x + y)$ up to and including terms of degree 4 in x and y . and also obtain an approximate value of $\ln(1.1)$.

ANSWER KEY

Q.No.	Ans.	Q.No.	Ans.	Q.No.	Ans.
1	A	8	B	15	C
2	B	9	A	16	A
3	A	10	A	17	A
4	A	11	B	18	B
5	A	12	C	19	B
6	C	13	A	20	A
7	B	14	B		